

Research Proposal

Provisional Title: Data Pipelines and Machine Learning for End-to-End Value-Chain Optimisation in Digitally Enabled Agricultural Ecosystems

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1.0 Motivation and Background

Global population growth continues to exert unprecedented pressure on global food systems. The world population increased from approximately 6 billion in 2000 to over 8 billion in 2025 and is projected to approach 10 billion by 2050, requiring an estimated 70% increase in global food production to meet rising demand (FAO, 2017; Ben and Hanana, 2021). These pressures are particularly acute in developing regions, where rapid population growth, climate volatility, infrastructural constraints, and fragmented agricultural supply chains pose significant threats to food security, economic stability, and rural livelihoods (Ogwu et al., 2024). In countries such as Nigeria, agriculture remains central to employment, income generation, and national resilience, amplifying the consequences of inefficiencies across agricultural value chains.

Agricultural innovation has historically progressed through mechanisation, the Green Revolution, and more recently, the emergence of digital agriculture (Pai and Iruku, 2026). Advances in modern farming technologies, including Internet of Things (IoT) sensors, remote sensing, automation, and artificial intelligence (AI)-driven analytics, have improved productivity and resource efficiency at the farm level. However, despite these gains, traditional agricultural value chains remain inefficient, opaque, and highly vulnerable to disruption. Persistent challenges such as seasonal variability, transportation bottlenecks, demand fluctuations, and quality control failures continue to drive post-harvest losses, increase costs, and undermine sustainability across production, storage, distribution, and retail stages (Alawode, 2025; Pai and Iruku, 2026).

Recent developments in machine learning (ML), big data analytics, and distributed computing offer powerful tools for addressing these challenges. Predictive models have been applied to crop yield estimation, disease detection, demand forecasting, logistics optimisation, and risk management. However, existing solutions are frequently developed as isolated applications, lacking the end-to-end data pipeline architectures required to integrate heterogeneous data sources and operationalise ML across the full agricultural value chain (Rozony et al., 2024). This fragmentation limits scalability, reduces responsiveness to emerging risks, and constrains the broader impact of ML-driven decision support.

Digitally enabled agricultural ecosystems such as integrating IoT sensing, solar-powered irrigation, cold storage, logistics tracking, digital marketplaces, and traceability technologies, present an opportunity to reimagine agricultural value chains from production to consumption. The HARVIST project provides a unique real-world context in which such integrated ecosystems can be piloted in Nigeria, embedding advanced analytics, machine learning, and blockchain-based traceability into operational agricultural systems. Crucially, the success of these ecosystems depends not only on individual ML models, but on the design of robust, scalable data pipelines capable of ingesting, harmonising, and analysing diverse data streams while supporting dynamic, predictive, and explainable ML across the entire value chain.

2.0 Research Aim and Objectives

2.1 Research Aim

The primary aim of this research is to design and evaluate an end-to-end, scalable data pipeline and machine learning framework that integrates heterogeneous, real-time agricultural data streams with advanced predictive and causal analytics to optimise agricultural value chains within digitally enabled ecosystems. Moving beyond static and survey-based analytical approaches, the framework will support continuous data ingestion, anomaly detection, and

explainable decision support for dynamic risk prediction and operational decision-making. Particular emphasis will be placed on resilience, system trustworthiness, and real-world deployment within African agricultural contexts.

2.2 Research Questions

1. How can real-time data pipeline architectures integrated with hybrid causal–predictive machine learning models improve resilience and efficiency in digitally enabled agricultural value chains?
2. What are the key integration challenges in harmonising heterogeneous agricultural data streams (IoT, logistics, storage, market, and climate data) within real-time operational environments?
3. How can the integration of fuzzy-set Qualitative Comparative Analysis (fsQCA/mIQCA) with XGBoost improve both causal interpretability and predictive generalisation in agricultural value-chain optimisation?
4. To what extent can anomaly detection techniques (e.g. Isolation Forest) enhance early identification of emerging disruptions that are not observable through static, survey-based datasets?
5. How effective is the proposed framework in improving operational performance, resilience, and sustainability within digitally enabled agricultural ecosystems in Nigeria?

2.3 Research Objectives

To achieve this aim, the research will pursue the following objectives:

1. Analyse agricultural data sources and integration challenges

Examine data generation, flow, and integration challenges across the agricultural value chain, from farm operations and IoT-enabled production to storage, logistics, and digital marketplaces, highlighting the limitations of survey-based datasets in capturing real-time, multi-source operational dynamics.

2. Design scalable, real-time data pipeline architectures

Design a modular data pipeline architecture capable of integrating heterogeneous agricultural data (e.g. IoT sensors, weather data, logistics tracking, cold storage monitoring, and market transactions), supporting both real-time stream processing and batch analytics for end-to-end visibility and responsiveness (Akanbi, 2024).

3. Develop hybrid predictive and causal analytics models

Develop supervised machine learning models using XGBoost for predictive analytics tasks such as yield estimation, demand forecasting, logistics optimisation, and risk prediction, while integrating fuzzy-set Qualitative Comparative Analysis (fsQCA) to capture configurational causal relationships and mitigate the limitations of purely data-driven prediction (Wu, et al., 2025).

4. Implement anomaly detection for emerging risk identification

Apply unsupervised learning techniques, particularly Isolation Forest (iForest, *as against Random Forest*), to detect anomalies and early warning signals across operational data streams, enabling proactive identification of disruptions, inefficiencies, and system failures that are not observable in static survey-based studies (Daramola et al., 2025).

5. Incorporate explainability and governance mechanisms

Integrate explainable AI techniques (e.g. SHAP values) to enhance interpretability, stakeholder trust, and regulatory alignment, ensuring that predictive and causal insights are transparent and actionable for farmers, extension workers, supply chain operators, and policymakers (Wu, et al., 2025).

6. Evaluate effectiveness and contextual applicability

Evaluate the proposed framework's impact on efficiency, resilience, and sustainability through a Nigeria-focused case study aligned with the HARVIST ecosystem, assessing scalability, governance considerations, and suitability for digitally enabled agricultural systems in developing economies.

3.0 Literature Review and Research Gap

3.1 Digital Agriculture and Smart Farming

Digital agriculture represents the latest stage in the evolution of agricultural innovation, following mechanisation and the Green Revolution, and is characterised by the use of IoT sensors, drones, satellite imagery, automation, and advanced analytics to improve productivity and sustainability (Pai and Iruku, 2026). Smart farming systems enable fine-grained monitoring of soil conditions, crop health, and environmental factors, supporting data-driven decision-making across irrigation, fertilisation, and pest control. Recent studies demonstrate that AI-driven soil monitoring and real-time sensing can significantly improve yield outcomes while reducing excessive fertiliser and water usage, thereby lowering environmental impact (Kumari et al., 2025).

Despite these advances at the farm level, smart farming solutions are often deployed in isolation, with limited integration into downstream value-chain activities such as storage, transportation, and market distribution. This disconnect constrains the ability of digital agriculture to deliver system-wide efficiency and resilience, particularly in developing-country contexts where infrastructural and coordination challenges persist (Pai and Iruku, 2026).

3.2 Machine Learning in Agricultural Supply Chains

Machine learning (ML) techniques have been widely applied across agricultural supply chains to support demand forecasting, inventory management, logistics optimisation, quality control, and risk management (Saxena et al., 2023). Supervised algorithms such as Random Forests, Support Vector Machines, Neural Networks, and Gradient-Boosting methods have shown strong predictive performance, while unsupervised techniques enable anomaly detection and pattern discovery in complex, high-dimensional datasets. These applications have contributed to reductions in post-harvest losses, improved route efficiency, and enhanced responsiveness to market fluctuations (Saxena et al., 2023).

However, much of the existing ML literature remains application-specific, focusing on discrete tasks such as yield prediction or route optimisation rather than addressing the agricultural value chain as an interconnected system. This fragmentation limits the capacity of ML solutions to capture cross-stage dependencies between production, storage, transportation, and consumption, and reduces their effectiveness in managing systemic risks and disruptions (Rejeb et al., 2025).

3.3 Data Pipelines and Big Data Integration in Agriculture

Agricultural data exhibits the defining characteristics of big data: volume, velocity, variety, veracity, and value, arising from heterogeneous sources such as IoT sensors, satellite imagery, logistics systems, weather feeds, and market platforms (Pai and Iruku, 2026). Distributed computing frameworks, including Hadoop and Apache Spark, have become standard tools for scalable data processing, yet their deployment requires substantial infrastructure and technical expertise, which can be prohibitive in resource-constrained environments (Rozony et al., 2024).

Ontology-based data integration approaches have been proposed to address semantic heterogeneity across agricultural datasets, while blockchain-based systems offer promise for traceability and trust. Nevertheless, ontology-driven frameworks struggle to scale in real-time operational settings, and blockchain applications in agriculture remain relatively immature, particularly with respect to performance, interoperability, and governance (Rozony et al., 2024). As a result, many ML-driven agricultural systems lack the robust, real-time data pipelines necessary to support continuous analytics and adaptive decision-making across the value chain.

3.4 Limitations of Existing Research Syntheses

Rejeb A, Rejeb K, and Hassoun A in 2025 provide a comprehensive topic-modeling-based review of ML applications in agricultural supply chains using Latent Dirichlet Allocation (LDA). Their study offers valuable high-level insights into dominant research themes and emerging application areas. However, several limitations are identified. First, while LDA is effective at uncovering broad thematic structures, it may overlook subtle interactions between closely related domains and fails to capture cross-domain dependencies within agricultural systems. Second, the approach is inherently static and descriptive, offering no operational or predictive capability. Third, the analysis is constrained by the selected literature corpus, limiting its generalisability. Finally, emerging paradigms such as explainable AI, edge computing, federated learning, and real-time analytics remain underexplored within their synthesis (Rejeb et al., 2025).

These limitations underscore a broader issue within the literature: the absence of frameworks that move beyond thematic classification toward operational, end-to-end system design capable of supporting real-world agricultural decision-making.

3.5 Research Gap

The reviewed literature reveals a significant and underexplored gap in current agricultural analytics research. Existing studies inadequately address how scalable, real-time data pipeline architectures can operationalise machine learning across entire agricultural value chains, particularly in developing-country contexts. *Most ML applications rely on static or survey-based datasets and focus on isolated optimisation tasks, limiting their capacity to support continuous monitoring, anomaly detection, and dynamic risk prediction in live operational environments.*

Wu et al. (2025) further demonstrate that static analytical approaches such as fsQCA, while effective for identifying configurational causal pathways, are insufficient for predicting emerging risk events. Their integration of fsQCA with XGBoost highlights the value of combining causal insight with predictive generalisation; however, *their framework remains constrained by survey-based data and does not address real-time, multi-source agricultural data streams or operational anomaly detection.*

Consequently, a clear research gap exists at the intersection of end-to-end data pipeline engineering, hybrid causal–predictive machine learning, and real-time agricultural value-chain optimisation. *Addressing this gap requires integrated frameworks that combine scalable data pipelines with supervised and unsupervised ML models to enable predictive analytics, anomaly detection, and explainable decision support within digitally enabled agricultural ecosystems.*

4.0 Research Methodology

4.1 Research Design

This research adopts a *design science and applied research methodology, integrating data pipeline architecture design, machine learning model development, and causal–predictive analysis* to optimise end-to-end agricultural value chains. The study moves beyond descriptive or survey-based approaches by focusing on operational, real-time, multi-source agricultural data streams and their use in predictive analytics, anomaly detection, and decision support.

A key methodological contribution is the adoption of *Machine-Learning Enhanced Qualitative Comparative Analysis (mlQCA)*, which integrates XGBoost-based machine learning with Qualitative Comparative Analysis (QCA). *This hybrid approach enables the study to overcome the traditional “small-N” limitation of QCA and to model complex causal configurations at scale, while retaining interpretability and causal reasoning.*

4.2 Data Sources

The study will utilise heterogeneous, real-world agricultural data sources spanning the full value chain, including:

- Farm-level IoT sensor data (soil moisture, temperature, humidity, irrigation status)
- Remote sensing data (drone and satellite imagery)
- Weather and climate data
- Cold-storage monitoring and equipment performance data
- Transportation and logistics data (GPS, RFID, delivery events)
- Market demand, pricing, and transaction data
- Traceability and transaction records (including blockchain-based systems where available)

This multi-source design addresses the limitations of survey-based datasets by enabling continuous, high-velocity, and context-aware analytics across production, storage, logistics, and markets.

4.3 Data Pipeline Architecture

A modular, scalable data pipeline architecture will be designed to support both batch and streaming ingestion, enabling real-time monitoring and historical analysis. The architecture will incorporate:

- Stream and batch processing frameworks (e.g. Apache Kafka and Spark-based architectures (Akanbi, 2024)).
- Cloud-based storage and compute infrastructure
- Semantic integration layers to address data heterogeneity
- Governance mechanisms for data quality, security, access control, and auditability

This pipeline design directly addresses the big-data integration challenges of volume, velocity, and variety identified in prior research (Pai and Iruku, 2026), while providing the operational foundation required for real-time ML deployment.

4.4 Machine Learning and Causal–Predictive Modelling

The analytical layer combines supervised, unsupervised, and causal–configurational approaches, ensuring both predictive performance and interpretability.

Supervised Learning: XGBoost

Extreme Gradient Boosting (XGBoost) will be employed for predictive tasks including crop yield estimation, demand forecasting, and supply-chain risk classification. XGBoost is selected for its ability to capture high-order feature interactions, manage noisy data, and deliver robust generalisation in complex agricultural datasets (Wu et al., 2025).

Unsupervised Learning: Isolation Forest

Isolation Forest will be used for anomaly detection to identify emerging risks not captured by labelled data, such as abnormal sensor behaviour, cold-chain failures, logistics delays, or market irregularities. This supports early warning and resilience by detecting deviations in real-time operational environments.

mlQCA Integration (Causal–Predictive Hybrid)

To address the limitations of purely predictive or purely configurational approaches, **the research integrates mlQCA (Machine-Learning Enhanced Qualitative Comparative Analysis)**. mlQCA enhances traditional QCA by embedding XGBoost and SHAP-based feature importance into the causal analysis process. Specifically, mlQCA will be used to:

- **Select conditions** using XGBoost and SHAP values to identify empirically important variables for causal modelling
- **Automate calibration** by deriving data-driven cut-off points from decision-tree splits, supporting fuzzy-set and crisp-set QCA

- **Iteratively optimise configurations**, testing thousands of condition combinations to identify high-coverage, high-consistency causal pathways
- **Improve truth-table quality** by reducing contradictions and enhancing configurational clarity

This approach directly addresses the limitations identified by Rejeb et al. (2025), where LDA-based reviews capture broad themes but fail to model subtle interactions or operational causality. It also extends the work of Wu et al. (2025) by embedding causal sufficiency logic **into real-time, multi-source data environments, rather than relying on static survey data.**

Explainability

SHAP values will be used across both XGBoost and mlQCA components to ensure transparency, interpretability, and stakeholder trust. This is critical for adoption by farmers, cooperatives, policymakers, and supply-chain actors.

4.5 Evaluation Strategy

The proposed framework will be evaluated across multiple dimensions:

- **Predictive performance** (accuracy, precision, recall, AUC)
- **Anomaly detection effectiveness** (detection latency, false-positive rates)
- **System performance** (latency, scalability, resource efficiency)
- **Causal consistency and coverage** within mlQCA configurations
- **Operational impact** on value-chain outcomes, including efficiency, resilience, and sustainability

Evaluation will be conducted using real and simulated datasets aligned with digitally enabled agricultural ecosystems, with a particular focus on applicability within Nigerian and wider African contexts.

Research Methodology Summary

This methodology advances the state of the art by:

- Moving beyond survey-based and static analyses
- Integrating **real-time data pipelines with hybrid causal–predictive ML**
- Embedding **mlQCA** to link interpretability, causality, and scalability
- Supporting anomaly detection and emerging risk prediction in live agricultural systems

5.0 Expected Contributions

This research is expected to make original contributions across methodological, technical, and applied dimensions.

Methodological contributions

The study advances agricultural ML research by integrating mlQCA with predictive machine learning (XGBoost) and anomaly detection (Isolation Forest) within an operational, real-time setting. By embedding causal configurational logic into scalable data pipelines, the research addresses a key limitation of existing studies that rely on static, survey-based or literature-driven analyses (Rejeb et al., 2025; Wu et al., 2025). The proposed framework demonstrates how causal sufficiency (fsQCA/mlQCA) and predictive generalisation (XGBoost) can be combined to support dynamic risk prediction and decision-making in complex agricultural systems.

Technical contributions

The research will design and evaluate a *modular, end-to-end data pipeline architecture* capable of integrating heterogeneous agricultural data streams (IoT, weather, logistics, storage, and market data) in both batch and streaming modes. This contributes practical design knowledge on how to operationalise ML across the entire

agricultural value chain, moving beyond isolated model development toward deployable, production-grade systems (Rozony et al., 2024; Pai and Iruku, 2026).

Applied contributions

From an applied perspective, *the study provides evidence-based insights into how predictive analytics, anomaly detection, and explainable AI can improve efficiency, resilience, and sustainability across agricultural value chains.* The findings are expected to inform practitioners, policymakers, and technology providers on the design of trustworthy, scalable, and context-aware digital agriculture solutions, particularly in developing-country contexts.

5.1 Relevance to the HARVIST Project and Nigerian Context

The proposed research is closely aligned with the objectives of the HARVIST (Hub for Agricultural Resilience through Value-chain, Irrigation, Storage, and Technology) project, which aims to pilot integrated, digitally enabled agricultural ecosystems in Nigeria. HARVIST's focus on solar-powered irrigation, cold storage, traceable logistics, predictive maintenance, and digital marketplaces requires not only individual analytics tools but a coherent data and ML backbone capable of linking these components into a unified ecosystem.

This research directly supports HARVIST by providing:

- A **scalable data pipeline framework** for integrating diverse operational data sources across farms, storage facilities, transport networks, and markets
- **Predictive models** for yield estimation, demand forecasting, and supply-chain risk identification
- **Anomaly detection mechanisms** for early identification of cold-chain failures, logistics disruptions, and equipment faults
- **Explainable decision support** to build trust among farmers, cooperatives, agribusinesses, and regulators

The Nigerian agricultural context presents unique challenges such as fragmented infrastructure, climate variability, data quality issues, and limited real-time visibility across value chains. By focusing on Nigeria as a primary case study, the research ensures contextual relevance and contributes actionable insights for digitally enabled agriculture in Sub-Saharan Africa. At the same time, the modular nature of the proposed framework supports scalability and replication across other developing regions, aligning with HARVIST's ambition for franchise-ready, extensible ecosystems.

6.0 Conclusion

This proposal addresses a critical gap in contemporary agricultural research: the absence of **end-to-end, operational frameworks** that integrate real-time data pipelines with dynamic, explainable machine learning across the full agricultural value chain. While prior studies have demonstrated the potential of ML for specific applications such as crop management or supply-chain optimisation, they remain fragmented, static, or disconnected from real-world deployment constraints (Rejeb et al., 2025; Saxena et al., 2023).

By combining **scalable data pipeline engineering, supervised and unsupervised machine learning, and mlQCA-based causal–predictive analysis**, this research offers a holistic approach to value-chain optimisation in digitally enabled agricultural ecosystems. The proposed framework not only enhances predictive accuracy and anomaly detection but also preserves interpretability, governance, and contextual applicability, key requirements for adoption in regulated, resource-constrained environments.

Aligned with the HARVIST project and grounded in the Nigerian context, the study aims to contribute both academically and practically to the development of **resilient, sustainable, and data-driven agricultural systems** capable of meeting the demands of a growing global population under increasing environmental and economic uncertainty.

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